Name_____ Date_____ Period_____

LESSON 3-5 · HOMEWORK PACKET · SHEET 1 OF 2

Comparing Ratios with Common Terms

Maryland Standard 6.RP.A.3b · Learning target – I can compare two ratios by scaling them until they share a common term.

REMEMBER

- To compare two ratios, scale them so one term is the **same**. Then compare the other terms.
- A bigger first number alone does **not** tell you which mix is stronger.

WORKED EXAMPLE

Punch A uses 2 cups juice to 3 cups water. Punch B uses 3 cups juice to 5 cups water. Which tastes more like juice?

Step 1 Match the juice (LCM of 2 and 3 is 6): A = 6 : 9, B = 6 : 10

Step 2 Same juice, and A has less water

Answer **Punch A tastes more like juice.**

PRACTICE

show your work · label units

1. Scale each ratio so the first term is 24.

a. 3 : 5 = 24 : ____

b. 4 : 7 = 24 : ____

c. 6 : 11 = 24 : ____

d. 8 : 9 = 24 : ____

2. Mix P uses 3 cups blue paint to 4 cups yellow. Mix Q uses 5 cups blue to 6 cups yellow. Complete the tables. Which mix is bluer?

P: blue	3	6	9	12	15
P: yellow	4				
Q: blue	5	10	15	20	25
Q: yellow	6				

Bluer mix_____

3. **Garden Seeds.** In Pack A, 9 out of every 12 seeds sprout. In Pack B, 13 out of every 16 seeds sprout.

a. Scale Pack A to 48 seeds: _____

b. Scale Pack B to 48 seeds: _____

c. Which pack sprouts better? _____

4. **Find the Error.** Drink X uses 4 spoons sugar to 9 cups water. Drink Y uses 6 spoons sugar to 15 cups water. Jamal says Y is sweeter because 6 > 4. Is he right? Explain.

_____

_____

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #___

AT HOME Mix two glasses of a drink with different amounts. Ask someone at home which is stronger and why.

Name_____ Date_____ Period_____

LESSON 3-5 · HOMEWORK PACKET · SHEET 2 OF 2

Comparing Rates with Benchmarks

Maryland Standard 6.RP.A.3b · Learning target – I can use a benchmark to compare two rates without dividing.

REMEMBER

- A **benchmark** is a friendly amount both rates can reach.
- Scale both rates to the benchmark, then compare. Same amount: **lower price** wins.

WORKED EXAMPLE

Which is the better buy: 4 cans for $3 or 6 cans for $4.80?

Step 1 Benchmark: 12 cans (4 × 3 and 6 × 2)

Step 2 12 cans: $3 × 3 = $9.00 vs. $4.80 × 2 = $9.60

Answer **4 cans for $3 is the better buy.**

PRACTICE

show your work · label units

1. Name a benchmark amount both rates can reach.

a. a 3-pack and a 4-pack ____ **b.** 6 minutes and 9 minutes ____

2. Kai types 90 words in 2 minutes. Lena types 140 words in 3 minutes. Circle the faster typist.

	Minutes	Words	Words in 6 minutes
Kai	2	90	
Lena	3	140	

3. **Bird Seed.** Store A: 5 pounds of bird seed for $8. Store B: 8 pounds for $12. Use a 40-pound benchmark.

a. Cost of 40 pounds: Store A ____ Store B ____

b. Which store is the better buy? _____

4. **Trail Mix.** Store C: 3 lb of trail mix for $7. Store D: 4 lb for $9. Use a 12-lb benchmark.

a. Cost of 12 pounds: Store C ____ Store D ____

b. Which store is the better buy? _____

5. **Reasoning.** Car 1 goes 150 miles on 5 gallons. Car 2 goes 200 miles on 8 gallons. Which is more efficient? Use a benchmark.

_____

_____

TEST PRACTICE

choose the best answer

6. Which benchmark could you use to compare 6 bagels for $4 and 8 bagels for $5?

A 14 bagels **B** 24 bagels **C** 16 bagels **D** 10 bagels

7. Dante walks 3 mi in 50 min. Ella walks 4 mi in 60 min. Using a 12-mile benchmark, who is faster?

A Dante (200 min) **B** Ella (180 min) **C** Dante (150 min) **D** Same speed

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #___

AT HOME Compare two sizes of one item in a store ad using a benchmark.